

PHYS 427: Lecture # 10

Cliff Sun

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Lecture Span

1. Chemical Potential

Chemical Potential

So far, $N = \text{const.}$ and $U = U_1 + U_2$, V and N are all const.

Diffusive equilibrium

$$dS = dS_1 + dS_2 \quad (1)$$

$$= \partial_{U_1} S_1 dU_1 + \partial_{V_1} S_1 dV_1 + \partial_{N_1} S_1 dN_1 \quad (2)$$

Then, $dU_1 = dU_2$ and etc. Moreover, $dS = dS_1 + dS_2 = 0$. Then, we arrive at

$$0 = \left[\frac{\partial S_1}{\partial U_1} - \frac{\partial S_2}{\partial U_2} \right] dU_1 + \dots \quad (3)$$

$$= \left(\frac{1}{T_1} - \frac{1}{T_2} \right) dU_1 + \left(\frac{p_1}{T_1} - \frac{p_2}{T_2} \right) dV_1 + \left(\frac{\mu_1}{T_1} - \frac{\mu_2}{T_2} \right) dN_1 \quad (4)$$

Here, we define the chemical potential μ as

$$\mu = -T \frac{\partial S}{\partial N} \quad (5)$$

Note, if $\mu < 0$, then the entropy increases w.r.t N . Then, we arrive at:

$$T_1 = T_2, \quad p_1 = p_2, \quad \mu_1 = \mu_2 \quad (6)$$

From eq. (2), we arrive that

$$dS = \frac{1}{T} dU + \frac{p}{T} dV - \frac{\mu}{T} dN \quad (7)$$

If N is allowed to vary, then this is the thermodynamic identity. Then, it follows that:

$$dU = T dS - p dV + \mu dN \quad (8)$$

We can then define

$$\mu = \frac{\partial U}{\partial N} \quad (9)$$

A difference in μ drives heat flow. If $\mu_1 > \mu_2$ with $T_1 = T_2$ and $p_1 = p_2$. Then,

$$dS = [\partial_{N_1} S_1 - \partial_{N_2} S_2] dN_1 \geq 0 \quad (10)$$

If the inside term is less than zero, then $dN_1 \leq 0$. Therefore, particles move from N_1 to N_2 (and therefore higher potential to lower potential).

Also,

$$\mu = \frac{\partial F}{\partial N} \quad (11)$$

Then,

$$dF = \left(\frac{\partial F_1}{\partial N_1} - \frac{\partial F_2}{\partial N_2} \right) dN_1 = 0 \quad (12)$$

Therefore, in equilibrium, we have that the inside term must be 0.

Example: Chemical potential of an ideal gas

We calculate

$$F = -k_B T \ln(Z), \quad Z = \frac{(n_Q V)^N}{N!} \quad (13)$$

Then,

$$F = -N k_B T (-\ln(n_Q V) + \ln(N) - 1) \quad (14)$$

Then, we calculate

$$\mu = k_B T \ln \left(\frac{n}{n_Q} \right) \quad (15)$$

For classical gas of $n \ll n_Q$, then $\mu < 0$. For argon, then $\mu = 0.42\text{eV}$.

Example

This is the Nernst equation. Two systems with a potential difference across each system. Then ideal gas with a charge Q . Then, $\epsilon_i = p_i^2/2m + qV_{1,2}$.

We calculate

$$Z = \frac{1}{N!} \left(\sum_i \exp(-\beta \epsilon_i) \right)^{N_1} \quad (16)$$

$$= \frac{1}{N!} (\exp(-\beta q v_1) n_Q V_1)^{N_1} \quad (17)$$

Note, we are ignoring interactions between particles. Then, you can calculate

$$F_1 = -N_1 k_B T (e^{-\beta q v_1} n_Q V_1) + N_1 k_B T \ln N_1 - N_1 k_B T \quad (18)$$

Note, the capital V is the volume, and lowercase v_1 is the potential. Then,

$$\mu_1 = -k_B T \ln \left(\frac{n_1}{n_Q} \right) + q v_1 \quad (19)$$

This is the internal chemical potential plus the external chemical potential. The Nernst equation is then

$$q \Delta v = k_B T \ln \left(\frac{n_2}{n_1} \right) \quad (20)$$

In general,

$$\Delta \mu_{\text{ext}} = -\Delta \mu_{\text{int}} \quad (21)$$

In dilute solutions,

$$\mu = \mu_{\text{sol}} + k_B T \ln \left(\frac{n_{\text{sol}}}{n_0} \right) \quad (22)$$

Here, n_0 is 1 mole of the solute. The gravitational potential is

$$\mu(h) = k_B T \ln(n(h)/n_Q) + mgh \quad (23)$$

Then,

$$\mu(h) = n(0) \exp(-mgh/k_B T) \quad (24)$$